Balancing method and experiment of a small spacecraft reaction wheel

W. De Munter, J. Lanting, T. Delabie, D. Vandepitte
KU Leuven, Department of Mechanical Engineering,
Celestijnenlaan 300, B-3001, Heverlee, Belgium

Abstract
Reaction wheels are considered as one of the main vibration sources on-board spacecraft. These vibrations can result in a poor pointing accuracy and/or stability and can therefore have an undesirable impact on the payload's output. A direct solution to minimize these vibrations is to balance the flywheel of the reaction wheel assembly. This paper elaborates the balancing process applied on one of the KU Leuven CubeSat reaction wheels. The results obtained by the balancing measurement and correction prototype set-up demonstrate that the initial static and dynamic unbalance can be reduced with respectively 83% and 23%, while the consistency and accuracy of the measurements should be increased by means of alternative measurements processing techniques.

1 Introduction

Small spacecraft are a booming class of satellites, with in particular the class of CubeSats. A one-unit (1U) CubeSat is a small satellite with standardized dimensions of $10 \times 10 \times 10\,\mathrm{cm}$, a mass of about $1\,\mathrm{kg}$, and a power consumption of approximately $1\,\mathrm{W}$. By combining several of these 1U cubes, derivatives such as the 2U, 3U and 6U CubeSats can be obtained. The main success of CubeSats can be assigned to this standardized concept in combination with the use of Commercial-Off-The-Shelf (COTS), leading to a reduction in development time and cost, and to affordable access into space [1].

The increasing amount of CubeSats and their applications goes together with an increase in system and pointing requirements. Stringent restrictions are imposed on the pointing budget and on the Attitude Determination & Control System (ADCS) to get high-quality data from these small platforms. The ADCS system determines the orientation - or attitude - of the spacecraft by means of different sensors, such as star trackers, gyroscopes, and/or magnetometers, and controls the orientation by means of different actuators, such as reaction/momentum wheels, magnetorquers, and/or thrusters.

The pointing accuracy and stability are determined by the ability of the ADCS to attenuate the different types of disturbances acting on the spacecraft. The environmental disturbance torques, for example, can be categorized as low-frequency disturbances ($< 1\,\mathrm{Hz}$), which can be corrected for by the ADCS. Reaction wheel and/or cryocooler vibrations, on the other hand, can be categorized as high-frequency disturbances ($> 1\,\mathrm{Hz}$), which require other strategies such as active/passive vibration isolation, dual-stage control approaches, and/or rotor unbalancing [2].

The Department of Mechanical Engineering at the KU Leuven develops an entire ADCS with in-house made reaction wheels for CubeSats. While the reaction wheels are the most accurate and agile actuators of the ADCS, they are also considered as one of the largest disturbance contributors on-board a spacecraft. Their disturbance profile can be estimated by means of the rotor dynamics theory and/or can be measured by means of characterization tests. The largest peaks in this profile can typically be assigned to the unbalances. One solution to reduce these peak magnitudes, and thus the vibrations of the reaction wheels, is to unbalance the reaction wheel's flywheel. Therefore, this paper will elaborate the unbalance procedure and test-setup applied on one of the KU Leuven CubeSat reaction wheels and discusses the experiment results.

The remainder of the paper is organized as such: section 2 explains the different definitions, representations and aspects of unbalances, section 3 describes the KU Leuven reaction wheel design and characterisation, section 4 demonstrates the potential impact of a balanced wheel on the pointing performance, section 5 discuss the balancing process approach, test set-up and test results, and section 6 concludes this paper.

2 Unbalances

A good understanding of the unbalance theory is recommended before initiating the balancing process aspects. This section therefore starts with the unbalance definition and its representation by means of equivalent masses and correction masses. Finally, the ISO standard unbalance tolerances and the unbalance estimations are mentioned.

2.1 Unbalance definition

According to the ISO-1940 standard [3], the unbalance is a condition which exists in a rotor when a vibration force or motion is imparted to its bearings as a result of centrifugal forces. Typically, two types of reaction wheel unbalances can be defined [4]: static unbalance and dynamic unbalance [1].

The static unbalance U_s, typically measured in $[\mathrm{g\,mm}]$, exists when there is a parallel shift between the rotation axis and the principal inertia axis of the rotor. This shift is called the eccentricity ϵ, typically given in $[\mu\mathrm{m}]$. The centre of mass (CoM) does not coincide with the geometric centre of the rotor. Given the total mass of the rotor m, The static unbalance U_s can be calculated according to equation 1a and is illustrated in Figure 1a.

$$U_s \overset{(a)}{=} m\,\epsilon \overset{(c)}{=} \widehat{m}_s\, r \overset{(e)}{=} (\widetilde{m}_s^A + \widetilde{m}_s^B)\, r \tag{1}$$

The dynamic unbalance U_d, typically given in $[\mathrm{g\,mm^2}]$, exists when there is an angle between the rotation axis and the principal inertia axis of the rotor. This angle χ is typically given in $[\mathrm{mrad}]$. The centre of mass (CoM) does coincide with the geometric centre of the rotor. Given the moment of inertia around the z-axis, the dynamic unbalance U_d can be calculated according to equation 2b and is illustrated in Figure 1b.

$$U_d \overset{(b)}{=} I_z\,\chi \overset{(d)}{=} 2\,\widehat{m}_d\, h_d\, r \overset{(f)}{=} (\widetilde{m}_d^A\, h_A + \widetilde{m}_d^B\, h_B)\, r \tag{2}$$

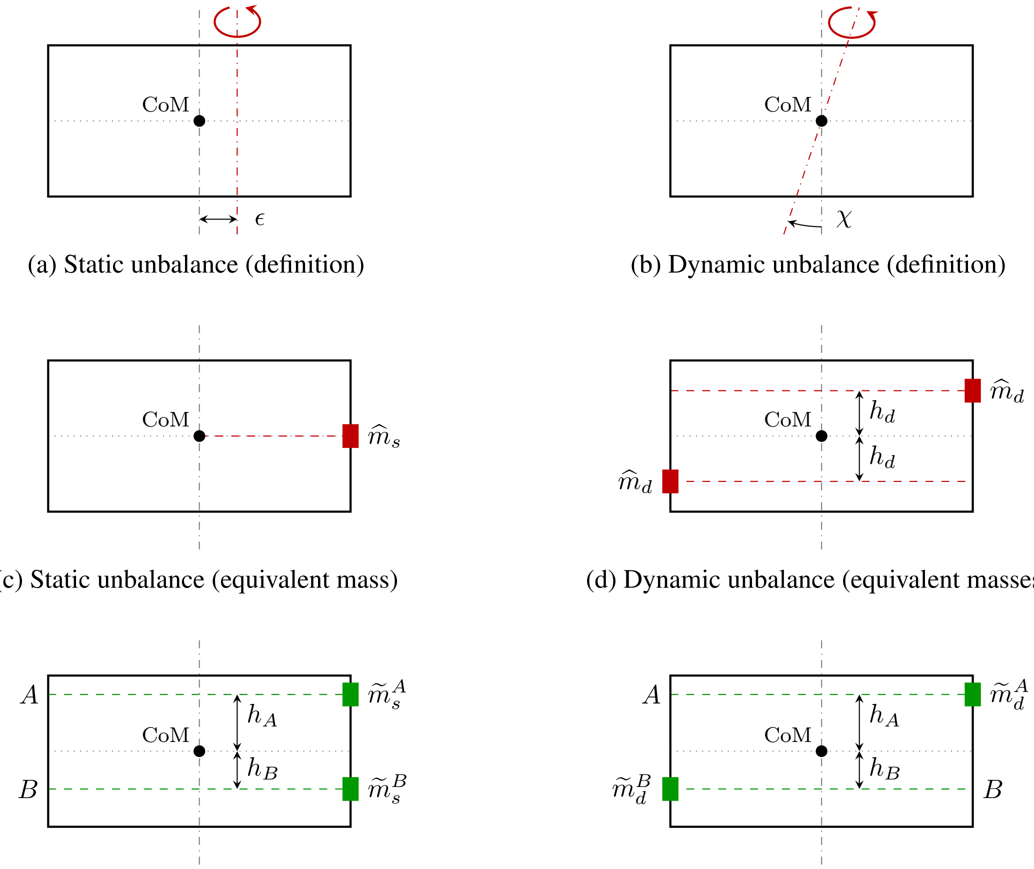
2.1.1 Equivalent masses

The unbalance definition in section 2.1 is useful to understand the physics and nature of the unbalances in rotating mechanism. However, for the balancing process, the representation by means of fictive point masses, so-called *equivalent masses* (EMs), is more meaningful. This representation assumes that the rotation axis is perfectly aligned with the mass moment of inertia I_z, with the EMs placed at a radial distance from the rotation axis, equal to the outer radius r of the rotor.

The static unbalance can be represented by one EM, $\widehat{m}_s$, on the plane perpendicular to the rotation axis and through the CoM, as illustrated in Figure 1c. The static unbalance can then be formulated according to equation 1c.

The dynamic unbalance can be represented by two EMs, $\widehat{m}_d$, on two different arbitrary planes perpendicular to the rotation axis with an equal axial distance h_d from the CoM, as illustrated in Figure 1d. Both EMs are positioned 180° with respect to each other. The dynamic unbalance can then be formulated according to equation 1d.

[1]According to the ISO-1940 standard [3] the two basic types of unbalances are the *static* and *couple* unbalance, while the combination or sum of these unbalances is defined as the *dynamic* unbalance. However, in reaction wheel unbalance literature the term couple unbalance is most of the time replaced by dynamic unbalance [5, 6]. The term *dynamic* unbalance in this paper therefore actually refers to the definition of the *couple* unbalance according to the ISO-1940 standard.



(a) Static unbalance (definition)

(b) Dynamic unbalance (definition)

(c) Static unbalance (equivalent mass)

(d) Dynamic unbalance (equivalent masses)

(e) Static unbalance (correction masses)

(f) Dynamic unbalance (correction masses)

Figure 1: Unbalance representations (rotor side view)

2.1.2 Correction masses

The static and dynamic unbalances can be compensated for by removing or adding *correction masses* (CMs). The magnitude and position of these CMs can be easily related to the representation by means of EMs as described in section 2.1.1. When removing masses, the CMs are equal to the EMs. When adding masses, the CMs have to be positioned 180° with respect to the EMs in the particular planes. In practice, however, the EM planes as illustrated in Figures 1c and 1d are not always accessible. In that case, at least two correction planes have to be defined. Figures 1e and 1f show two arbitrarily planes A & B with respectively a height h_A & h_B with respect to the CoM plane. These heights does not have to be equal neither they have to be related to the equivalent dynamic unbalance height ($h_A \neq h_B \neq h_d$). (Note that the CMs illustrated in Figures 1e and 1f representing the case where the CMs are *removed* to compensate for the equivalent masses illustrated in respectively Figures 1c and 1d.) When these correction planes A & B are defined together with their heights h_A & h_B, and the unbalances U_s & U_d are measured, the CMs for both planes for both unbalances can be calculated according to equations (3) - (6).

$$\widetilde{m}_s^A = U_s\, h_B / [r\,(h_A + h_B)] \tag{3}$$

$$\widetilde{m}_s^B = U_s\, h_A / [r\,(h_A + h_B)] \tag{4}$$

$$\widetilde{m}_d^A = U_d / (2\, r\, h_A) \tag{5}$$

$$\widetilde{m}_d^B = U_d / (2\, r\, h_B) \tag{6}$$

2.2 Unbalance tolerances

For some applications, the rotating mechanism design is driven by vibration limits determined by experimental evaluation or analysis. These vibration limits are then translated into unbalance tolerances.

On the basis of worldwide experience and similarity considerations, balance quality grades G have been established which permit a classification of the balance quality requirements for typical machinery types. The ISO 1940 standard [3] provides guidelines to select unbalance tolerances based on the rotor/machine type and the maximum wheel speed Ω_{max} with units $[\mathrm{rpm}]$. The balancing quality grade G with units $[\mathrm{mm/s}]$ is a value that then can be applied to calculate the maximum tolerated eccentricity ϵ_{max} with units $[\mathrm{mm}]$ according to equation 7.

$$\epsilon_{max} = G/\Omega_{max} \tag{7}$$

2.3 Unbalance estimation

For some other applications, the rotation mechanism design is not driven by vibration limits. In those cases the maximum unbalances can be estimated according equations 1a and 2b as described in section 2.1. The mass m and I_z can be derived from the rotor CAD model, while the maximum eccentricity ϵ_{max} and maximum angle χ_{max} can be derived from the series of worst case bearing and dimension tolerances of the total assembly. Furthermore, Monte Carlo simulations can estimate the maximum unbalance in the rotor wheel itself due to a non-uniform mass distribution. Together, these estimated maximum unbalances will give already an order of magnitude and can be used to validate characterisation tests of the particular rotating mechanism.

3 KU Leuven Reaction Wheel

The balancing process described in section 5 in this paper will be applied on one of the KU Leuven reaction wheels (RWs). This section therefore describes the current design of the RW first, as it will impose some specific restrictions and adaptations during the balancing process. Furthermore, this section discusses the results of a characterisation test of one of the KU Leuven RWs, as it will clarify the unbalance measurement method as described in section 5.

3.1 Design

The KU Leuven reaction wheel design exists of a rotating flywheel (rotor) which is suspended by two radial ball bearings positioned in an upper and lower housing respectively (stator), as illustrated in Figure 2 [7]. The flywheel is driven by a brushless DC motor via a compact Oldham coupling. The reaction wheel assembly has dimensions of approximately $40 \times 40 \times 28\,\mathrm{mm}$ and a total mass of $80\,\mathrm{g}$.

Within its dimensions, the KU Leuven RW design was optimized to have a maximum torque and maximum angular momentum capacity. While, the maximum torque directly derives from the brushless DC motor performance, the angular momentum capacity was maximized by augmenting both the flywheel's inertia I_z and the potential rotational speed Ω. The RW bearing fits and radial play were therefore selected to mainly minimize the friction, and not necessarily to minimize the unbalance. Following this design rule, the worst case unbalances can be estimated as described in 2.3. Together with the main flywheel parameters, the maximum estimated errors and maximum estimated unbalances are listed in Table 1. The KU Leuven RW was designed to have a maximum rotational speed of $10\,000\,\mathrm{rpm}$. Together with the maximum estimated error from Table 1, a balancing grade of $G = 40\,\mathrm{mm/s}$ is obtained. This grade corresponds to the class of *car wheels* and *car drive shafts* [3]. A balancing grade of $G = 0.4\,\mathrm{mm/s}$ corresponds to the class of *gyroscopes*, which are more similar devices to RWs, and would lead to a maximum tolerated unbalance of $\epsilon_{max} = 0.5\,\mu\mathrm{m}$. This number is, however, hard to achieve with the current existing and affordable balancing tools. Instead of imposing this stringent unbalance tolerance on the design in order to limit the vibrations,

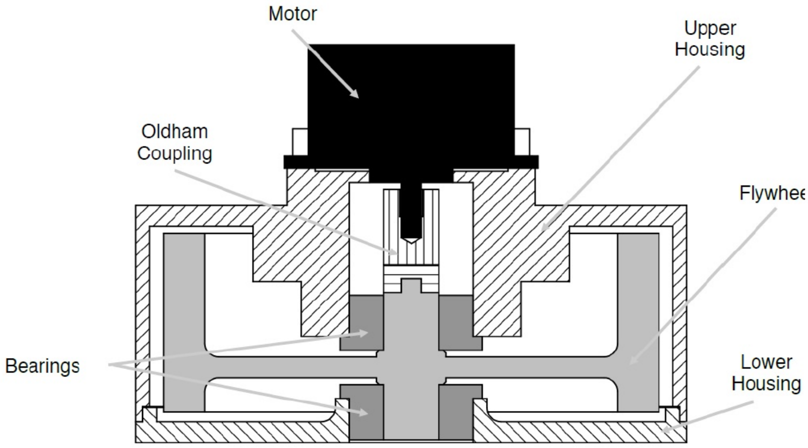


Figure 2: Cutout of the KU Leuven reaction wheel design [7]

Table 1: The maximum estimated KU Leuven RW design unbalance values [8]

Parameter	Maximum error	Maximum unbalance
$m = 40\,\mathrm{g}$	$\epsilon_{max} = 40\,\mu\mathrm{m}$	$U_{s,max} = 1.6\,\mathrm{g\,mm}$
$I_z = 9.6 \times 10^{-6}\,\mathrm{kg\,m^2}$	$\chi_{max} = 6.2\,\mathrm{mrad}$	$U_{d,max} = 60\,\mathrm{g\,mm^2}$

the rotational speed of the RW can be controlled to maintain a mean rotational speed around for example $2000\,\mathrm{rpm}$ during high-pointing manoeuvres [2].

3.2 Characterisation

All rotating mechanisms inherently have unbalances due to limited tolerances as explained in section 2. On top of these unbalances, rotating mechanisms have a more complex disturbance behaviour, especially when they incorporate ball bearings. These components have multiple bodies rotating at different rotational speeds producing lower and higher order of harmonics. Moreover, the rotating mechanism is a mechanical structure with its own eigenfrequencies which can potentially amplify these unbalances and/or harmonic disturbances. The field of *rotor dynamics* can help to predict the total disturbance behaviour of a rotating mechanism theoretically. Alternatively, characterisation tests can help to quantify the total disturbance behaviour experimentally. During a characterisation test, the dynamics (forces & moments) of the rotating mechanism running at a certain rotational speed are measured. When repeating this measurement over the entire range of rotational speeds of the particular mechanism, so-called *waterfall plots* can be derived which visually provide insight in the disturbance behaviour. Characterisation tests can be performed to validate the theoretical model of the rotating mechanism [5, 10], to verify the health status of the rotating mechanism based on the harmonic disturbances [6], or to quantify the unbalances for balancing processes [11]. Figure 3 shows the power spectral density (PSD) of the force F_x for a rotational speed of $3000\,\mathrm{rpm}$, derived from a characterisation test on one of the KU Leuven RWs. From this figure it is possible to distinguish three main contributors to the total force disturbance:

1. The **static and dynamic unbalances** originate from the unequally divided mass of the rotor and the limited tolerances of the assembly. If both the assembly is designed - and the rotor is mounted in a proper way (aligned bearings without excessive radial play), these unbalances are easily recognized in

[2]The RWs on-board a spacecraft are typically maintained within a certain rotational speed range by means of a technique called *momentum dumping* where other type of actuators such as magnetorquers or thrusters will gradually provide the necessary torque to the spacecraft [9].

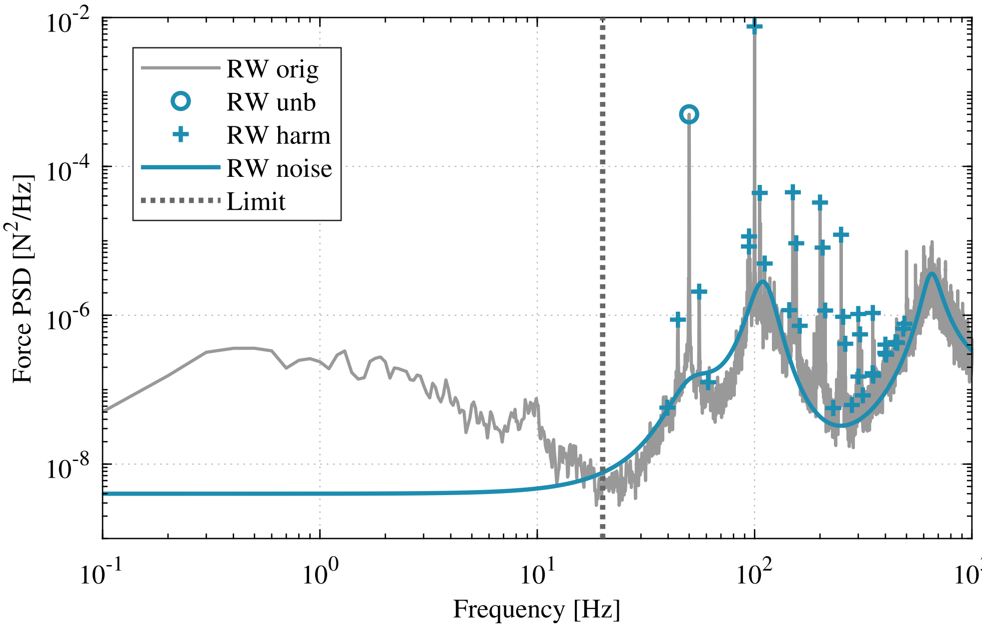


Figure 3: Power spectral density (PSD) of force F_x for a rotational speed of $3000\,\mathrm{rpm}$ (measurements are only valid above the seismic noise frequency limit of $20\,\mathrm{Hz}$)

the waterfall plots as a straight line with large amplitudes and in the spectral density plots as a large spike, as illustrated in Figure 3 (RW unb).

The corresponding unbalance forces and moments are periodic functions with a frequency equal to the wheel speed $f = h\Omega$ $(h = 1)$ and an amplitude proportional to the wheel speed squared Ω^2. If treated with care, the large spikes originating from the unbalance disturbances can be avoided or attenuated to minimize their impact on the pointing performance.

2. The **harmonic disturbances** originate from the ball bearing components. The harmonic numbers h_i can be estimated beforehand based on the dimensions of the ball bearing components. Furthermore, harmonic numbers with large amplitudes can indicate improper mounting of the rotor in the assembly. The harmonic disturbances are recognized in the waterfall plots as straight lines over the frequency spectrum and in the spectral density plots as small spikes, as illustrated in Figure 3 (RW harm).

   The corresponding harmonic forces and moments are periodic functions with a frequency equal to the harmonic times the wheel speed $f_i = h_i\Omega$ $(h_i \neq 1)$ and an amplitude which is defined to be proportional to the wheel speed squared Ω^2. However, when these harmonic lines intersect with one of the structural modes of the RW assembly, the amplitude function will deviate from this assumption.

3. The **broadband noise** is the remaining part of the disturbances, originating from the bearing components and the motor working principle (ripple, cogging, ...). For simulation purposes, the noise level can also be approximated as illustrated in Figure 3 (RW noise). The broadband noise is actually the largest contributor to the pointing performance, due to the area under the broadband noise curve representing the noise variance.

   The corresponding noise forces and moments are treated as white noise. Its magnitude can change over the frequency spectrum, as the RW assembly can be assumed as a mass-damper-spring system with lateral and radial eigenmodes, amplifying the noise level.

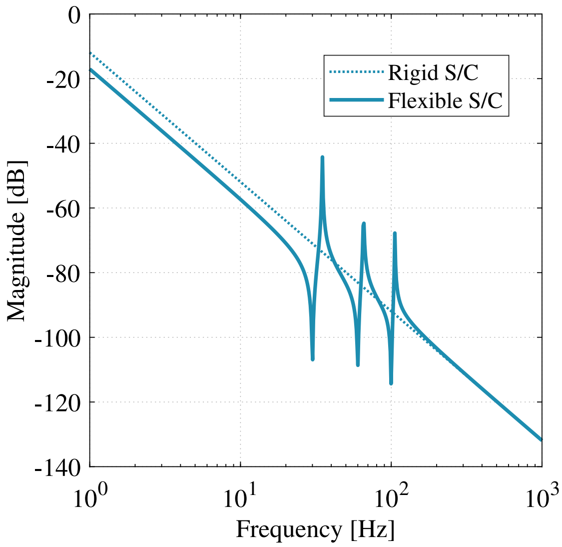


Figure 4: Frequency response of a rigid and flexible spacecraft model

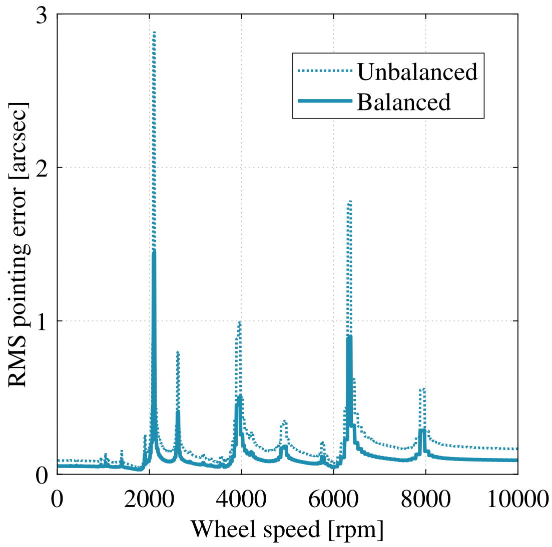


Figure 5: Pointing response due to the (un)balanced RW

4 Balancing motivation

Although the reaction wheel noise is the largest contributor to the pointing performance budget [12], the spikes originating from the unbalance forces and torques can have a significant impact as well, especially in the frequency range above 10 Hz. While section 1 briefly explained the motivation for balancing qualitatively, this section will motive the balancing process quantitatively.

The ADCS systems on-board small spacecraft typically have a bandwidth in the range of 0.1 Hz to 1 Hz, being capable of attenuate disturbances up to these frequencies. Higher frequency disturbances are therefore passed through the spacecraft structure. Typically, the spacecraft rigid body is represented as a double integrator plant of which the frequency response is illustrated in Figure 4. In reality however, the structure will show some eigenmodes with low damping at higher frequencies, resulting in a flexible spacecraft model. The numerical values of the flexible spacecraft model elaborated in this section can be found in Appendix A.

When a reaction wheel mounted in the spacecraft is running at a rotational speed corresponding to one of the structural eigenmodes of the spacecraft, a large RMS pointing error will be observed, as illustrated in Figure 5. For example, the unbalance forces and torques of a reaction wheel running at around 2100 rpm will be amplified by the first structural eigenmode at 30 Hz, resulting in a RMS pointing error of 3 arcsec. When this reaction wheel would be balanced by a factor of two, the resulting RMS pointing error will reduce with the same factor due to linearity. It can therefore be concluded that a reduction in unbalances will reduce the RMS pointing error with the same factor.

In addition to actively reducing the unbalances by means of a balancing process, the large spikes originating from the unbalance forces and torques can also be avoided passively. One option is to avoid running the RW at those wheel speeds that coincide with the spacecraft's eigenmodes. This condition can not always be guaranteed, however, as the reaction wheel speed will change continuously to stabilise the spacecraft. An alternative option is to properly design RW dampers to attenuate the unbalance forces in the desired frequency region.

In the end, it is up to the RW manufacturer and/or the space mission engineers to decide which options are preferred.

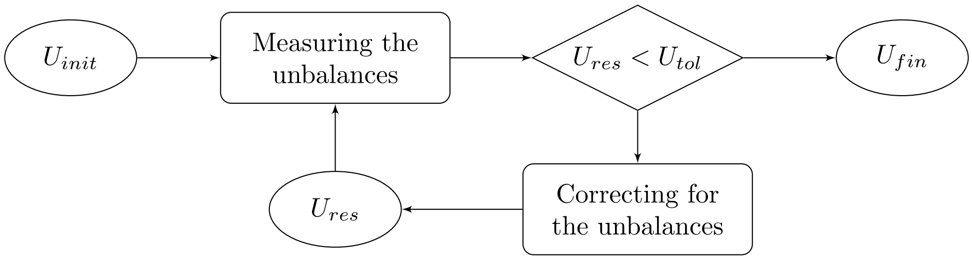


Figure 6: Overview of the balancing process

5 Balancing process

The balancing process exists of two main actions: measuring and correcting for the unbalances, as illustrated in Figure 6. Once manufactured, *initial* unbalances U_{init} exists in the rotating mechanism. These initial unbalance have to be measured first in order to decide whether to initiate the correction action or to end up with the *final* unbalances U_{fin}. When a correction is applied, the *residual* unbalances U_{res} have to be measured to decide the correction action have to be repeated or not. The balancing process illustrated in Figure 6 is valid for both the static and dynamic unbalances.

First the balancing approach, incorporating the measuring and correction operations, is discussed in section 5.1, secondly the balancing test set-up and operation is discussed in section 5.2. Finally, the balancing results and improvements are discussed and evaluated in section 5.3 and 5.4.

5.1 Balancing approach

There exists several approaches to measure and correct for the unbalances during the balancing process. The two main approaches for each balancing aspect are listed in Table 2 and are discussed in this section.

Regarding the measuring action, both the magnitude and angle of the static and dynamic unbalance vector have to be measured. These parameters will respectively determine to what extend and where to correct for the unbalances. Regarding the correction action, the number of correction planes, the correction operation and the correction level have to be decided. These parameters mainly depend on the particular mechanism to be balanced.

- The unbalance vector magnitudes can be derived (kinematically) from acceleration data obtained by means of accelerometers mounted on the rotor. Alternatively, the magnitude of the unbalance vectors can also be derived (dynamically) from force data obtained by means of a dynamometer.
- The unbalance vector angles can be measured directly or indirectly. In the latter case, a magnitude measurement is executed before and after a small calibrated mass is mounted on the rotor under a predefined radial angle. Based on the second reference measurement, the angle of the initial unbalance can be derived. Alternatively, the angle can be measured directly by means of an optical encoder.
- The number of correction planes depends on the magnitude and distribution of the initial unbalance as well as on the rotor's design. In case only the static unbalance should be corrected for, one correction plane is sufficient; otherwise two or more correction planes are necessary.
- The correction operation can be done by adding or removing the correction masses.
- The correction level defines the level on which the correction is applied. At rotor level, only the rotor component is balanced, while at assembly level, the rotor suspended by its bearings in the final set-up is balanced.

For the balancing process applied on the KU Leuven RW, the second approach for each balancing aspect from Table 2 was chosen. A dynamometer underneath the RW measures the forces and torques, while an optical encoder measures the angles. Correction masses are removed from two correction planes for each unbalance at assembly level [13, 11].

Table 2: Balancing approaches

Balancing aspect	First approach	Second approach
Measurement of magnitudes	Kinematically	Dynamically
Measurement of phases	Indirectly	Directly
Correction operation	Adding	Removing
Correction planes	One	Two or more
Correction level	Rotor level	Assembly level

5.2 Balancing test set-up and operation

This section describes the first test setup prototype used for the balancing process on the KU Leuven RW. The first two subsections elaborate the measurement set-up and processing respectively, while the next two subsections elaborate the correction set-up and parameters respectively.

5.2.1 Measurement set-up

The KU Leuven RW was placed on a Kistler dynamometer (type 9119AA1) to measure the forces and torques as illustrated in Figure 7. This dynamometer consists of four 3-component piezo-electric force sensors. The current signals of these force sensors are further converted into a proportional voltage signal by a Kistler multichannel charge amplifier (type 5070A). The Kistler dynamometer was rigidly mounted on a Newport RS2000 air table to isolate the dynamometer from ambient seismic vibrations ($< 10\,\mathrm{Hz}$). The same measurement set-up was used in [8] and allows to measure the RW disturbance forces and torques from which the unbalance magnitudes can be derived.

The unbalance phases are derived from optical encoder measurements. The encoder pattern is illustrated in Figure 8 and has two rings. The larger inner ring exists of a pattern with 24 black and white segments of $15°$ each. The three consecutive black segments on the flywheel are used as the reference position (R). The smaller outer ring exists of a pattern with 360 black and white segments of $1°$ each. The inner ring is used in combination with the optical sensor, the outer ring is used for positioning the flywheel during the correction action. The encoder pattern was fixed on the top ridge of the reaction wheel's flywheel, inside the housing components.

The optical sensor is a light reflection switch with a high photosensitive receiver for short distance, operating in the infrared range [14]. Since the flywheel is completely covered by the housing components in the original RW design, a hole had to be foreseen in the upper housing part to get the encoder in the view of the sensor, as illustrated in Figure 7. The sensor was mounted on a fixed lever arm close to the encoder patter in order to guarantee the maximum sensor's sensitivity.

Both the dynamometer and optical sensor measurements are collected by one data acquisition (DAQ) system to ensure they are synchronized, which is a critical requirement for determining the unbalance phases. Furthermore the RW wheel speed is controlled to be constant during each measurement to avoid phase differences and frequency leakage.

5.2.2 Measurement processing

In order to determine the unbalance magnitudes, the original time series from the dynamometer are converted to the frequency domain first. Next, the largest peak is searched for in a narrow frequency range around the rotational speed frequency, corresponding to the unbalance force or torque. By repeating this measurement for different wheel speed, the measurement accuracy can be increased by averaging and/or fitting.

In order to determine the unbalance phases, both the original time series from the dynamometer and optical sensor are filtered first by means of a narrow band-pass filter. This filter has a central frequency placed at

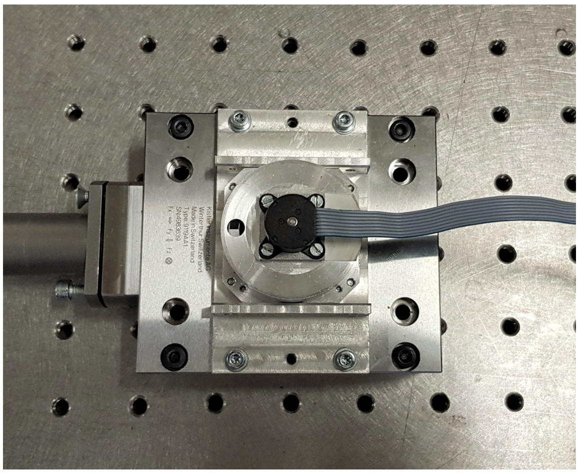


Figure 7: The KU Leuven RW mounted on the dynamometer

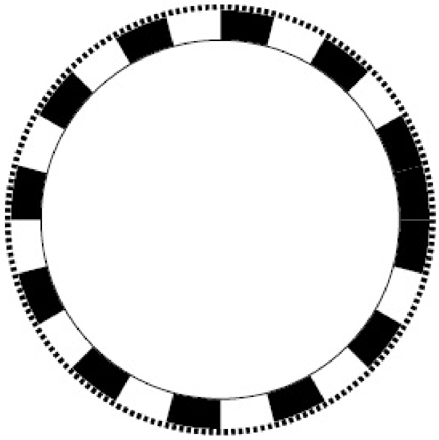


Figure 8: The optical encoder

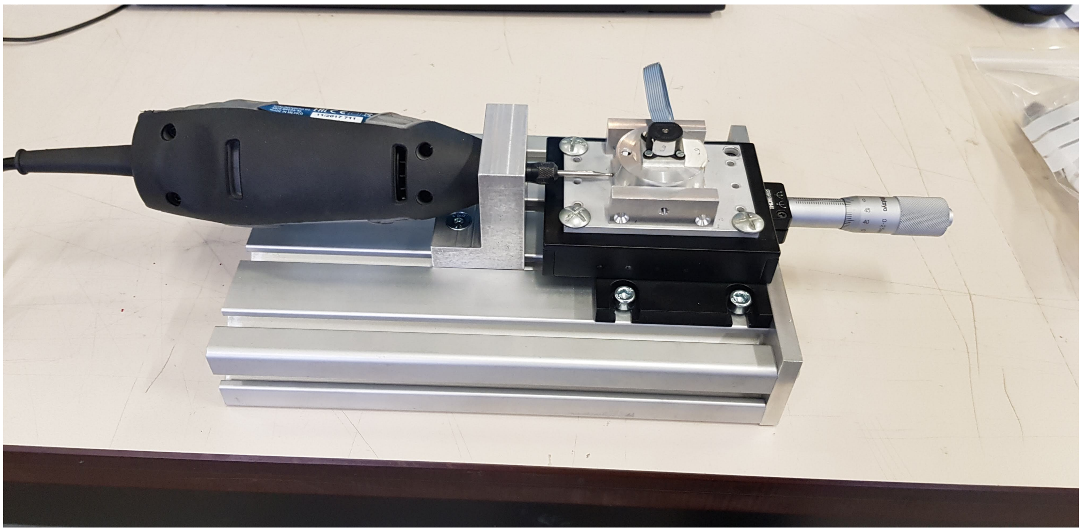


Figure 9: The correction set-up with the drilling tool and translation stage

the rotational wheel speed $f_c = \Omega/60$ and has a frequency width of $\Delta f = 4\,\mathrm{Hz}$. The filter is imposed on both signals to apply the same phase shift introduced by the filter. Next, the largest peak values are tracked in both filtered time series and are compared to each other to determine the phase differences.

5.2.3 Correction set-up

During the correction step, correction masses will be removed by means of a drilling tool, as illustrated in Figure 9. The drilling tool with a $2\,\mathrm{mm}$ diameter spherical diamond head is clamped on a portable aluminium base together with a $10\,\mathrm{\mu m}$-resolution translating stage, on which the reaction wheel assembly can be mounted. Two holes, corresponding to two user-defined correction planes, were made in the upper housing part of the reaction wheel assembly to reach the flywheel with the drilling head. Furthermore, two more holes were made on both sides of the upper housing part which allow to clamp the flywheel with some screws during the drilling process. This correction set-up allows to precisely remove the necessary correction masses.

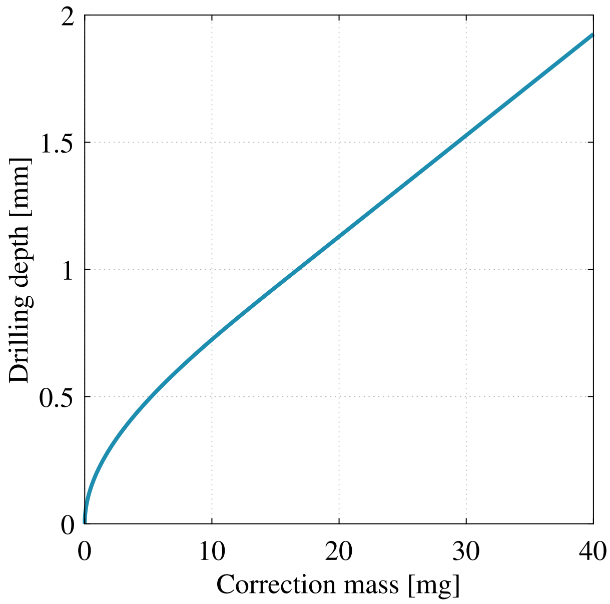


Figure 10: The correction drilling depth $\widetilde{x}$ in function of the correction mass $\widetilde{m}$

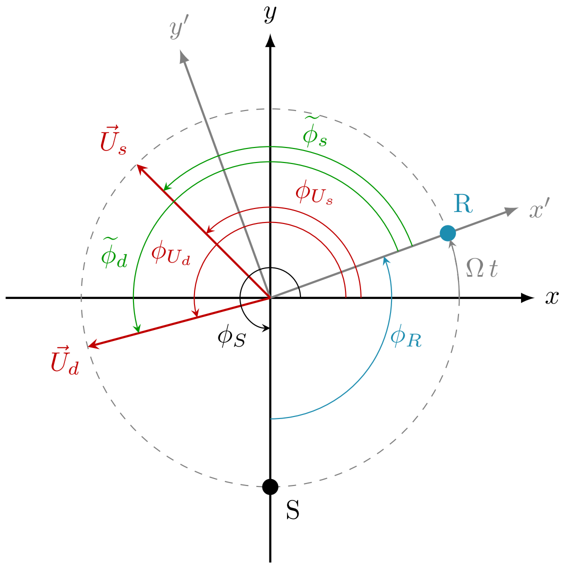


Figure 11: The phase definitions

5.2.4 Correction parameters

Once the magnitudes of the unbalance vectors are determined, the correction masses for both correction planes can be calculated according to equations 3 - 6 as discussed in section 2.1.2. Next, these correction masses have to be translated into correction depths $\widetilde{x}$ to be drilled. The relation between a correction mass and a correction depth is formulated in equation 8 and illustrated in Figure 10, taking into account the material density of steel $\rho = 8050\,\mathrm{kg\,m^2}$ and the drilling head radius $R = 1\,\mathrm{mm}$.

$$\widetilde{V} = \frac{\widetilde{m}}{\rho} = \begin{cases} \frac{\pi\widetilde{x}^2}{3}\,(3R - \widetilde{x}) & \text{if } \widetilde{V} \leq \frac{2\pi R^3}{3} \\ \frac{\pi R^2}{3}\,(3\widetilde{x} - R) & \text{if } \widetilde{V} > \frac{2\pi R^3}{3} \end{cases} \tag{8}$$

$$\begin{cases} \widetilde{\phi}_s = [\phi_{U_s}(t) - \phi_R(t)] - \phi_S & (9) \\ \widetilde{\phi}_d = [\phi_{U_d}(t) - \phi_R(t)] - \phi_S & (10) \end{cases}$$

The correction phase angles $\widetilde{\phi}_s$ and $\widetilde{\phi}_d$ can be calculated according to equations 9 and 10 respectively and are illustrated in Figure 11. In this figure, the black coordinate system (x-y-z) represents the static measurement frame of the dynamometer. The gray coordinate system (x'-y'-z') represents the frame fixed on the rotating flywheel with its x-axis aligned with the reference point (R), i.e. the three consecutive black segments on the optical encoder. The sensor position (S) is fixed with respect to the dynamometer frame and defined by the user-defined static angle ϕ_S. Depending on the applied dynamometer disturbance data x or y, this angle should be defined with respect to the corresponding axis. The static sensor position angle ϕ_S, together with the phase difference $[\phi_R(t) - \phi_{U_s}(t)]$ between the reference signal and the force/torque signal, allows to determine the correction phase angles.

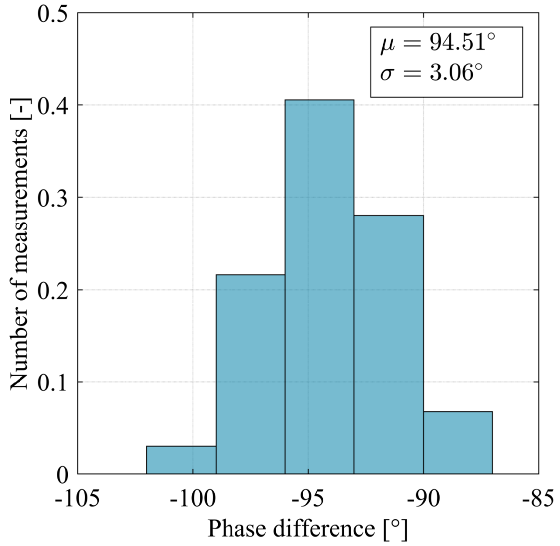
Once the correction drilling depths $\widetilde{x}_s^A$, $\widetilde{x}_s^B$, $\widetilde{x}_d^A$ and $\widetilde{x}_d^B$, and the correction angles $\widetilde{\phi}_s$ and $\widetilde{\phi}_d$ are determined, the correction can be executed.

5.3 Balancing results

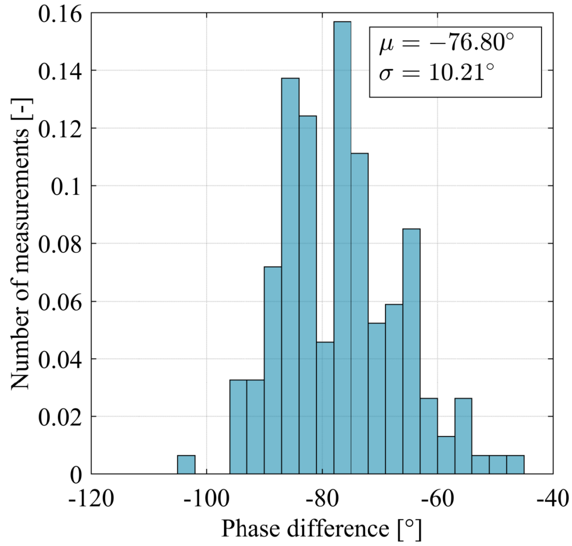
The unbalance magnitudes and phases were determined based on the measured time series for six different wheel speeds for one KU Leuven RW as listed in Table 3. Since the phase difference can be measured for

Table 3: The measured static [$\mathrm{g\,mm}$] and dynamic [$\mathrm{g\,mm^2}$] unbalance magnitudes and the normal distribution parameters (μ, σ) of the measured unbalance phases [°]

Speed	Unbalance magnitude				Unbalance phase							
Ω [rpm]	U_{s_x}	U_{s_y}	U_{d_x}	U_{d_y}	μ_{F_x}	σ_{F_x}	μ_{F_y}	σ_{F_y}	μ_{T_x}	σ_{T_x}	μ_{T_y}	σ_{T_y}
1219	0.06	0.04	7.59	3.52	-76.80	10.21	157.48	25.37	57.70	2.92	138.48	4.56
1610	0.06	0.04	7.08	3.55	-82.45	4.33	-165.27	13.99	56.79	2.36	152.63	4.20
1971	0.08	0.05	7.92	2.99	-96.87	5.37	-168.78	6.32	50.92	4.46	156.91	5.09
2253	0.11	0.04	10.10	1.46	-94.51	3.06	-151.61	14.60	58.57	2.95	137.96	16.76
2524	0.12	0.04	9.70	3.61	-83.16	5.14	142.15	11.32	77.51	6.36	99.88	10.50
2632	0.14	0.04	11.39	4.69	-75.96	7.01	128.77	23.52	85.17	9.85	78.36	11.26



(a) $\Omega = 2253\,\mathrm{rpm}(F_x)$



(b) $\Omega = 1219\,\mathrm{rpm}(F_x)$

Figure 12: Histograms of the phase difference measurements

each period of the signal, multiple measurements can be derived to obtain the expected normal distribution, characterised by its mean μ and standard deviation σ. This distribution is illustrated in Figure 12 for a measurement with a narrow and wide standard deviation respectively. The bin width of these histograms was set at $6\,\Omega/f_s$ [°] rounded to the first integer. For a wheel speed of $\Omega = 2253\,\mathrm{rpm}$ and a sampling frequency of $f_s = 4096\,\mathrm{Hz}$, the bin width becomes 3°. This number is equal to the resolution at which the phase difference is determined. For high wheel speeds, this resolution becomes too low to determine the unbalance phases accurately.

The results shown in Table 3 are clearly not consistent. It was noticed that the wheel speed controller could not stabilize the speed sufficiently. Therefore these results are difficult to interpret. Nevertheless an unbalance correction was performed to see if a difference could be noticed. The initial and residual (or final) unbalance values for this correction are listed in Table 4. These results show that it is possible to reduce the static and dynamic unbalance with 83% and 23% respectively with the current test-setup.

5.4 Balancing set-up and operation improvements

Section 5.3 clearly illustrated that the current measurements are not sufficiently consistent to execute an unbalance correction accurately. Therefore the following improvements, related to the difficulties encountered during the measurement and correction actions, are proposed:

Table 4: Initial and residual unbalance values respectively before and after balancing

	U_s [g mm]	ϕ_s [°]	U_d [g mm^2]	ϕ_d [°]
Initial	0.239	249	10.1	232
Residual	0.041	148	7.8	195

- An important requirement for the balancing process is to ensure a very stable wheel speed. A fine-tuned wheel speed controller can easily solve this issue..
- Multiple measurements on different wheel speeds will be required to increase the accuracy and reliability of the unbalance correction by means of statistics.
- The phase difference measurement resolution is too low at the moment. However, the sample frequency of $4096\,\mathrm{Hz}$, which determines the resolution in the time domain, can not be increased further. An alternative would be to derive the phase difference from the frequency domain, where the resolution is determined by the total measurement time, which can be increased.
- The correction set-up should allow to perform the corrections in both correction planes without disconnecting the drilling tool and/or reaction wheel. Furthermore, the drilling setup could be combined with the measurement set-up to reduce the disconnection steps even more. In this case, the total setup should be characterised first to ensure the lowest eigenfrequencies are higher than the region of interest ($500\,\mathrm{Hz}$).
- The current correction set-up does not take care of the removed material. The drilling chips can therefore nest in the bearings, which can reduce the reaction wheel's life time significantly. An extraction system could solve this issue.

6 Conclusion

This paper introduced the idea to balance CubeSat reaction wheels to reduce the micro-vibrations in these small satellites and therefore increase their pointing stability. First, the different representations and aspects of unbalances were explained. Next, the KU Leuven reaction wheel design was discussed to gain insight of the particular rotating mechanism to be balanced, where after the balancing process was motivated by means of a flexible spacecraft linear analysis to validate the potential positive impact of balancing reaction wheels. Subsequently, the balancing process, with in particular the measurement and correction set-up, was explained. Finally, the first test results were discussed and analyses. These results demonstrate that the current measurement set-up allows to reduce the initial unbalances. Further improvements will increase the consistency and accuracy of the test setup.

Acknowledgements

Strategic Basic (SB) PhD fellow at Research Foundation - Flanders (FWO).

References

[1] R. Shimmin, "Small spacecraft technology state of the art," NASA Ames Research Center, Tech. Rep., 2015.

[2] M. Hasha, "High-performance reaction wheel optimization for fine-pointing space platforms: Minimizing induced vibration effects on jitter performance plus lessons learned from hubble space telescope

for current and future spacecraft applications," in *Proceedings of the 43rd Aerospace Mechanisms Symposium*, NASA Ames Research Center, 2016, pp. 373–400.

[3] *Mechanical vibration - Balance quality requirements for rotors in a constant (rigid) state - Part 1*, International Standard Organisation Std. ISO 1940-1:2003(E), 2003. [Online]. Available: https://www.dcma.mil/Portals/31/Documents/NPP/Forms/ISO_1940-1.pdf

[4] G. Genta and F. F. Ling, *Dynamics of Rotating Systems*. Springer, 2005, ch. 3, pp. 93–103.

[5] R. A. Masterson, D. W. Miller, and R. L. Grogan, "Development and validation of reaction wheel disturbance models: Empirical model," *Journal of Sound and Vibration*, vol. 249, no. 3, pp. 575–598, 2002.

[6] L. M. Phuoc, "Micro-disturbances in reaction wheels," Ph.D. dissertation, TU Eindhoven, 2017.

[7] T. Delabie, "Star tracker algorithms and a low-cost attitude determination and control system for space missions," Ph.D. dissertation, KU Leuven, 2016.

[8] W. De Munter, T. Delabie, and D. Vandepitte, "Characterization of reaction wheel micro-vibrations," in *Proceedings of the 2018 International Conference on Noise and Vibration Engineering (ISMA)*, Leuven, Belgium, 2018, pp. 3511–3526.

[9] M. J. Sidi, *Spacecraft Dynamics and Control - A Practical Engineering Approach*. Cambridge University Press, 1997, ch. 7, pp. 189–194.

[10] D. Kim, "Micro-vibration model and parameter estimation method of a reaction wheel assembly," *Journal of Sound and Vibration*, vol. 333, pp. 4214–4231, 2014.

[11] D.-I. Cheon, E.-J. Jang, and H.-S. Oh, "Reaction wheel disturbance reduction method using disturbance measurement table," *Journal of Astronomy and Space Sciences*, vol. 28, no. 4, pp. 311–317, 2011.

[12] J. Vandersteen, J. Vennekens, and R. Walker, "Nano- to small satellite pointing errors: Limits of performance," in *Proceedings of the 2016 Small Satellites Systems & Services (4S) Symposium*, Valletta, Malta, 2016.

[13] T. Winters, "Ontwikkeling van een balanceermethode voor een cubesat reactiewiel," Master's thesis, KU Leuven, 2020.

[14] *Opto Interrupter*, Everlight Electronics Co., Ltd., 2008, rev. 1.

Appendix

A Flexible spacecraft model and RW disturbance model

This appendix elaborates both the flexible spacecraft model as well as the reaction wheel disturbances that are used in section 4 to illustrated the impact of RW disturbances on the pointing performance. The spacecraft model is formulated according to equations 11 - 14, which are derived from the spacecraft model diagram as illustrated in Figure 13. Based on the parameters listed in Table 5, the stiffness coefficients $k_i = (2\,\pi\, f_i)^2\, I_i$ and damping coefficients $c_i = 4\pi\, f_i\, I_i\, \zeta_i$ can be calculated.

$$\begin{aligned} I_0\dot{\omega}_0 = {} & T + k_1(\theta_1-\theta_0) + c_1(\omega_1-\omega_0) \\ & + k_2(\theta_2-\theta_0) + c_2(\omega_2-\omega_0) \\ & + k_3(\theta_3-\theta_0) + c_3(\omega_3-\omega_0) \end{aligned} \tag{11}$$

$$I_1\dot{\omega}_1 = -k_1(\theta_1-\theta_0) - c_1(\omega_1-\omega_0) \tag{12}$$

$$I_2\dot{\omega}_2 = -k_2(\theta_2-\theta_0) - c_2(\omega_2-\omega_0) \tag{13}$$

$$I_3\dot{\omega}_3 = -k_3(\theta_3-\theta_0) - c_3(\omega_3-\omega_0) \tag{14}$$

The reaction wheel disturbance model parameters are listed in Tables 6 - 8. The unbalance values listed in Table 6 are the initial unbalances of the *unbalanced* wheel of Figure 5, while these values were divided by a factor two to simulate the impact of the *balanced* wheel in Figure 5.

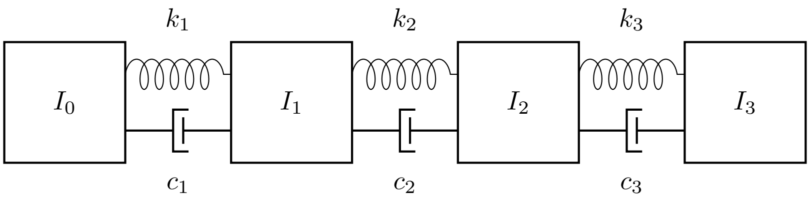


Figure 13: Flexible spacecraft model diagram

Table 5: Flexible spacecraft parameters

Inertia [kg m^2]		Eigenfrequencies [Hz]		Damping [%]	
I_0	0.10	-	-	-	-
I_1	0.05	f_1	30	ζ_1	0.2
I_2	0.02	f_2	60	ζ_2	0.5
I_3	0.01	f_3	100	ζ_3	0.3

Table 6: RW unbalance simulation parameters

Ω [rpm]	U_s [g mm]	U_d [g mm^2]
$10 \ldots 10\,000$	0.5	40

Table 7: RW noise simulation parameters

S_F [N/$\sqrt{\text{Hz}}$]	S_T [Nm/$\sqrt{\text{Hz}}$]
10^{-10}	10^{-12}

Table 8: RW harmonic simulation parameters
(with C_{F/T_i} defined as a percentage of the unbalance values from Table 6)

i	1	2	3	4	5	6	7	8	9	10	11
h_i [-]	0.8	1.0	1.1	1.5	1.9	2.0	2.2	3.0	4.1	5.0	7.2
C_{F/T_i} [%]	0.15	1.00	0.08	0.07	0.03	0.08	0.05	0.01	0.02	0.08	0.10